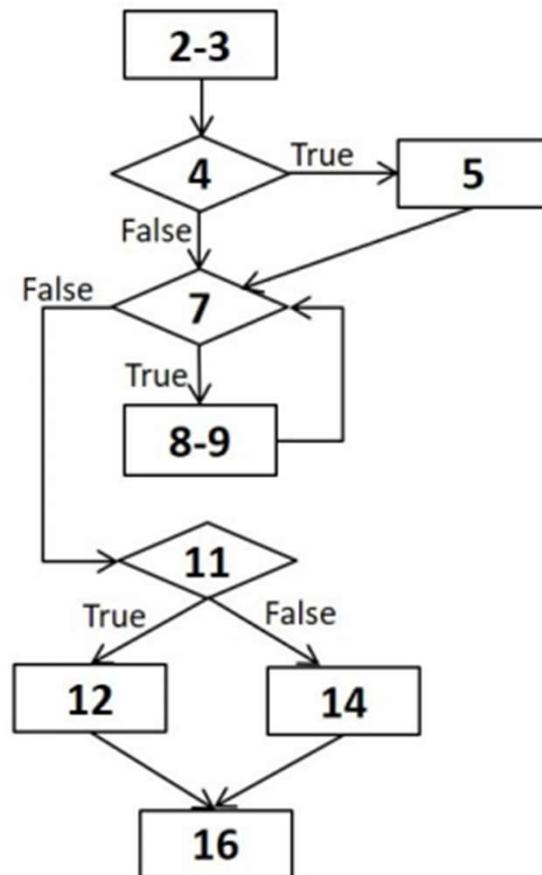


Tutorial 9: Control Flow Testing

Answer 1(a): Control flow graph for `sum(int n, int upperbound)`



Answer 1(b): Cyclomatic complexity

Edges = 11

Nodes = 9

Cyclomatic complexity = $11 - 9 + 2 = 4$

Answer 1(c): Test cases coverage

100% Statement coverage:

Path#1: 2-3, 4, 5, 7, 8-9, 7, 11, 12, 16

Path#2: 2-3, 4, 5, 7, 8-9, 7, 11, 14, 16

Test cases:

Path#1: $n = -1$, upperbound = 1; expected outcome: "The sum is 1"; 1

Path#2: $n = -1$, upperbound = 0; expected outcome: "The sum is too large!"; 1

100% Branch coverage:

Path#1: 2-3, 4, 5, 7, 8-9, 7, 11, 12, 16

Path#2: 2-3, 4, 7, 8-9, 7, 11, 14, 16

Test cases:

Path#1: $n = -1$, upperbound = 1; expected outcome: "The sum is 1"; 1

Path#2: $n = 1$, upperbound = 0; expected outcome: "The sum is too large!"; 1

Answer 1(d): Linearly independent paths, test case and expected outcome

Path#1 (baseline): 2-3, 4, 7, 11, 12, 16

Path#2: 2-3, 4, 5, 7, 11, **14**, 16

Path#3: 2-3, 4, 7, 8-9, 7, 11, 12, 16

Path#4: 2-3, 4, 7, 11, 14, 16

Test cases:

Path#1: $n = 0$; upperbound = 1; expected outcome: "The sum is 0"; 0

Path#2: $n = -1$; upperbound = -1; expected outcome: "The sum is too large!"; 0

Path#3: $n = 1$; upperbound = 1; expected outcome: "The sum is 1"; 1

Path#4: $n = 0$; upperbound = -1; expected outcome: "The sum is too large!"; 0